

Joey Jaehyeok Kim

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Education

Georgia Institute of Technology
B.S. Mechanical Engineering

Jun 2022 – Jul 2026 (Expected)
GPA: 3.78/4.00

Experience

Tesla Lathrop, California
Test Engineer Jan 2025 – Jul 2025

- Diagnosed and troubleshot electronic subassembly failures through test program execution, data analysis, and fault injection
- Achieved 98% FPY and 10x mating cycle by designing pogo-pin harness connectors with increased ergonomics.
- Improved end-of-line FPY by 4% by building and integrating an organizational system across all general assembly lines
- Cut expenses by \$20,000 by developing ESD-safe fixtures for electronic components to prevent arc flashes and shorting
- Partnered with maintenance, process, and quality engineers to consistently increase production rate and quality
- Led supplier interface and technical documentation to ensure fixture manufacturability and reliability

Cognosos Atlanta, Georgia
Mechanical Engineer Jan 2024 – Aug 2024

- Generated comprehensive and precise 2D schematics and assembly drawings from 3D models
- Incorporated DFM principles to optimize CAD for efficient and cost-effective manufacturing of plastic enclosures
- Created and executed Design Validation Tests to assess the functionality and durability of a multipurpose duress button
- Managed a backlog of 300+ units of 3D prints and regularly serviced the QIDI X-Max 3D printer
- Validated CAD drawings of products to ensure accuracy, completeness, and compliance with industry standards
- Oversaw the prevention of data loss when transferring SOLIDWORKS files onto Onshape, and Arena PLM and QMS

Computational Design and Metamaterials Lab Atlanta, Georgia
Research Assistant under Dr. Bolei Deng Aug 2024 – Present

- Validating TPMS models through Instron shear and axial testing to correlate experimental stress-strain data with simulation outcomes
- Creating a differentiable simulation that can optimize metamaterial structure to replace padding in motorcycle headgear
- Performed material testing and data analysis to correlate simulation predictions with real-world deformation behavior
- Designed a tube with both dilative and compressive properties as an alternative to surgical ETTs
- Modeled bistable unit cells to create metamaterials with 2 degrees of freedom on flat surfaces
- Evaluated metamaterial design to achieve neutral buoyancy and allow 3 degrees of freedom underwater

Projects

TPMS for Headgear Impact Protection Sep 2025 – Present

- Differential simulation based on Cauchy momentum equation and Hooke's law to compute strain-energy gradients to inverse design metamaterial structures with optimized 3D cell geometry
- Achieved 7% greater shear strength and improved strength-to-weight efficiency through FEA-driven comparisons between conventional helmet foam and TPMS lattice liners
- Simulating thermal heat transfer through TPMS geometries to analyze airflow and temperature regulation at transient and steady states
- Maintain $HIC < 250$ and $BrIC < 0.45$ and meet DOT impact requirements through iterative tuning of lattice parameters

Voice-Controlled Robotic Arm Assistant Jul 2025 – Present

- Designed a 6-DOF robotic arm with CAD and integrated AI voice assistant (LISA).
- Implemented motion control via Python with logging and unit tests for reliability.

Modeling Equity Option Performance Under Stochastic Dynamics Aug 2025 – Sep 2025

- Black-Scholes pricing with Greeks sensitivity analysis and a Monte Carlo engine to estimate hit probabilities, expected payoff timing, and fair value vs. market price
- Integrated live market data via Yahoo Finance API
- Interactive Tkinter GUI with ticker autofill, input validation, and valuation verdicts

Detecting Faulty PCBs using Thermal Image Analysis May 2025 – Jun 2025

- Detect faulty connections on PCBs using Gaussian filtering, morphological operations, and region-based analysis
- Automated report generation with annotated visuals and defect metrics

Autonomous Robot Design and Fabrication Aug 2023 – Dec 2023

- Designed and fabricated an autonomous robot with a custom drivetrain and a pulley-adapted scissor lift system
- Utilized DC motors, IR sensors, solenoids, AC adapters, potentiometers, and pneumatics to diversify robot capabilities

Skills

- **Software:** SOLIDWORKS, Vector CANape, LabView, Siemens HMI, CATIA, Onshape, AutoCAD, Rhino, Arena, Cura, PrusaSlicer, Qidi Print, ROS, Arduino, JobControl, Inkscape, BarTender, TeraTerm, WavePad, VideoPad, Kibana
- **Programming:** Go, Swift, TP Programming, Java, Python, MATLAB, C++
- **Technicals:** FEA, FMEA, DFM, GD&T, Soldering, 3D print, laser cut, woodwork, lathe, bandsaw, drill press